(NOTE, changes on details are to be expected)
Title: Self-made machine vision with extra steps for Pikwel.
Desc.: Making a modular Machine vision AI from somewhat from scratch, then distilling it to acceptable size so it runs on x (to be determined) hardware. End result should be two artifacts: an AI's that's plug-and-play and a grant AI that can be used to teach smaller AIs. Also camera system's effect on the machine vision. Company also wishes to keep some part of the research hidden.

Methodology: Design science.

Research goals (to be extended):
* (Main) Can "self-made" AI system be implemented as plug-and-play machine vision with modular components while allowing them to run on x (to be determined) hardware systems.
* How much resources must be used while using such AI's and how to optimize them.
* How different camera systems affect the AI's capability to perform

Tools used:
* OpenCV, PyTorch Vision, TensorFlow, Scikit-image (Machine vision and AI libraries)
* visual studio code (main coding tool)
* GitHub (version control)
* TurboQuant (Vector Quantization)

Timetable:

Name	Desc.	Date/deadline
Planed	Publish work plan	2-6-2026
Official Start	Lets do this. On site meeting with Pikwel	10-6-2026
Foundation	Write methodology and related research	20-6-2026
The interesting part	Implement, test, and document. Doing the actual research.	14-7-2026
Finalize document	Ensure that everything is as it should be	30-7-2026
Realization	Sudden but planned realization that something went wrong.	31-7-2026
Recover	Fix everything in a panic and summer heat.	14-8-2026
The meeting	Present everything done during summer to the guiding professor.	??-8-2026
Salary	They start to pay salary for this	1-9-2026
Redo	Lets redo this. Once more with feeling.	1-10-2026
Done	Finished, ready to be evaluated	1-12-2026
Graduate	I have did done it. I have become the software engineer in a 4,5 years.	6-1-2027
Kela money stops	Kela does pay no more. If the progress goes here, the situation just escalated to an awful level.	1-3-2027
"*Life happens*"	If everything falls apart, this is my absolute last deadline. This is the last plan that should **NOT** happen.	1-6-2027

The plan has been set. The path is lit. Now I only require the strength to follow it.

Related Works:

A research called "*TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate*", introduces a methodology to Vector Quantization which aims to compress high dimensional vectors. Problem with compression usually is the loss of data but this research claims that their TurboQuant methodology is major improvement to preserving important vector data while compressing. The research provides formal proof that TurboQuant is close to the best information-theoretically achievable distortion rate which a Vector Quantizer could do. (Zandieh *et al.*, 2025)

A book called "*Hands-On ML Projects with OpenCV : Master Computer Vision and Machine Learning Using OpenCV and Python*" has excellent examples, tutorials, and guides how to make different machine visions. The book focuses on OpenCV (Open Source Computer Vision Library) is a free library that includes optimized algorithms for real-time image processing, machine learning, and computer vision, and it has python library for machine vision (*OpenCV: OpenCV modules*, no date). (Mugesh, 2023)

A book called "*Machine Vision Technology*" introduces machine vision as improvement upon human vision. Humans aren't capable to detect minor millimeter differences while machine vision can achieve precision of 0.001 mm or even higher. The book goes over how machine vision works in fine detail. (Chen and Chen, 2026)

References:

Chen, B. and Chen, S. (2026) *Machine Vision Technology*. Singapore: Springer Nature Singapore (Advanced and Intelligent Manufacturing in China). Available at: https://doi.org/10.1007/978-981-95-1184-6.

Mugesh, S. (2023) *Hands-On ML Projects with OpenCV : Master Computer Vision and Machine Learning Using OpenCV and Python*. 1st edn. Orange Education PVT Ltd. Available at: https://ebookcentral.proquest.com/lib/lut/detail.action?docID=30682670 (Accessed: 2 June 2026).

OpenCV: OpenCV modules (no date). Available at: https://docs.opencv.org/4.13.0/ (Accessed: 2 June 2026).

Zandieh, A. *et al.* (2025) 'TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate'. arXiv. Available at: https://doi.org/10.48550/arXiv.2504.19874.